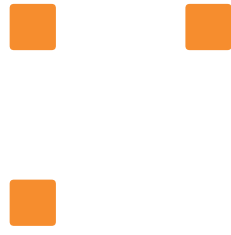


Decision Trees

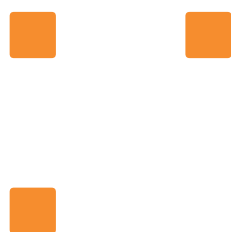


For classification and regression.

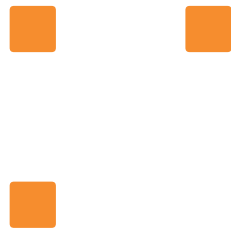


Outline

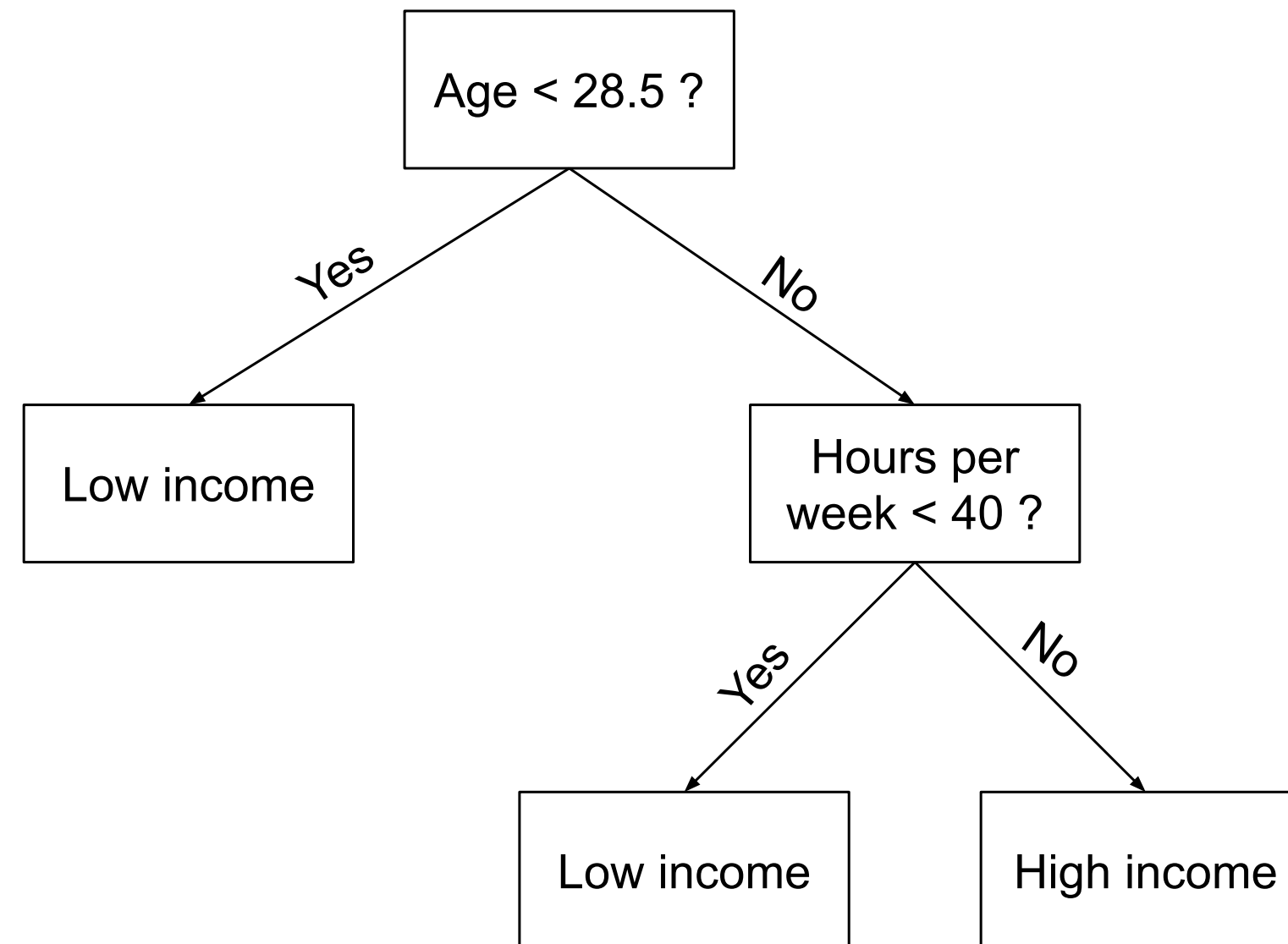
- What is a decision tree?
- Training trees for classification & regression
- Impact of the tree depth on overfitting

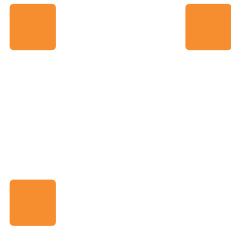


What is a decision tree?

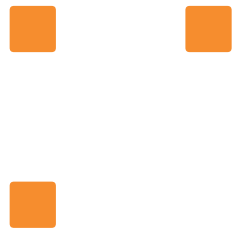


Example of a decision tree

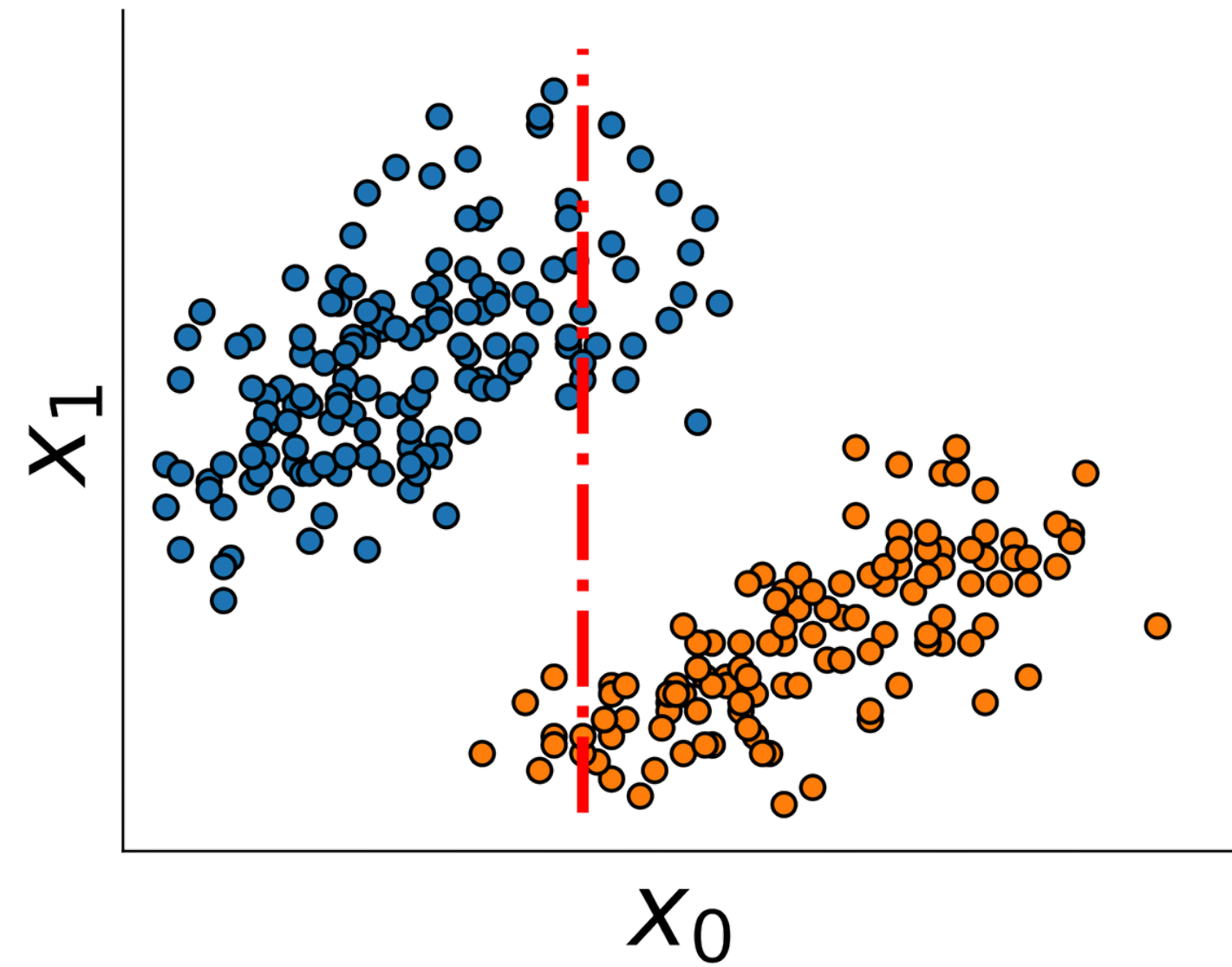
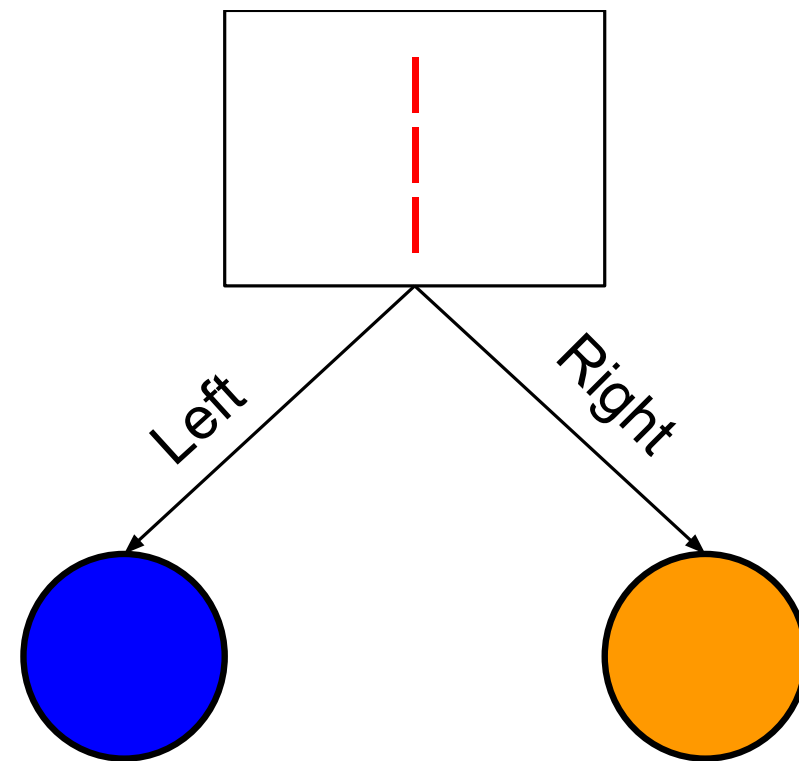


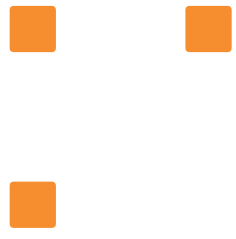


Decision trees in classification

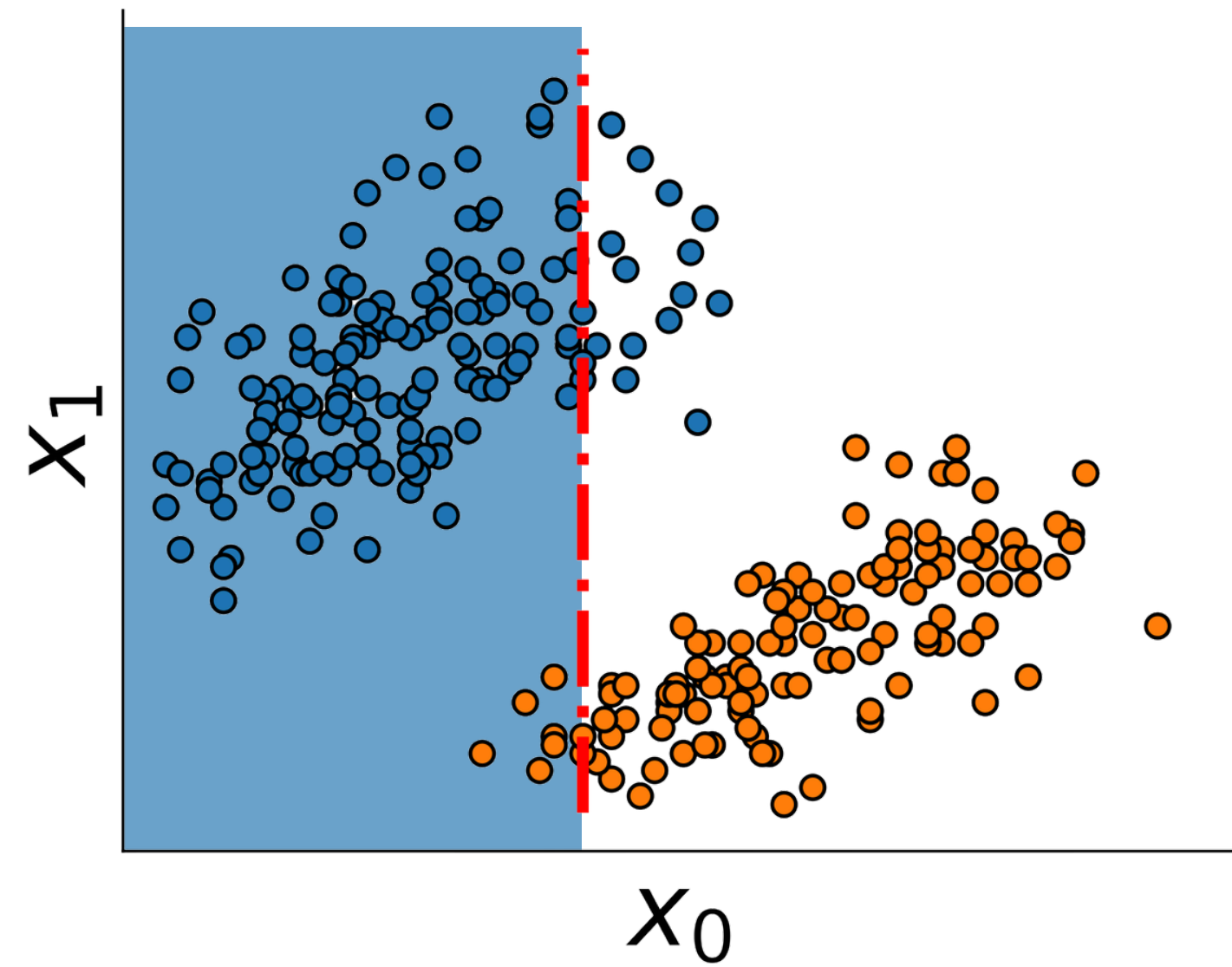
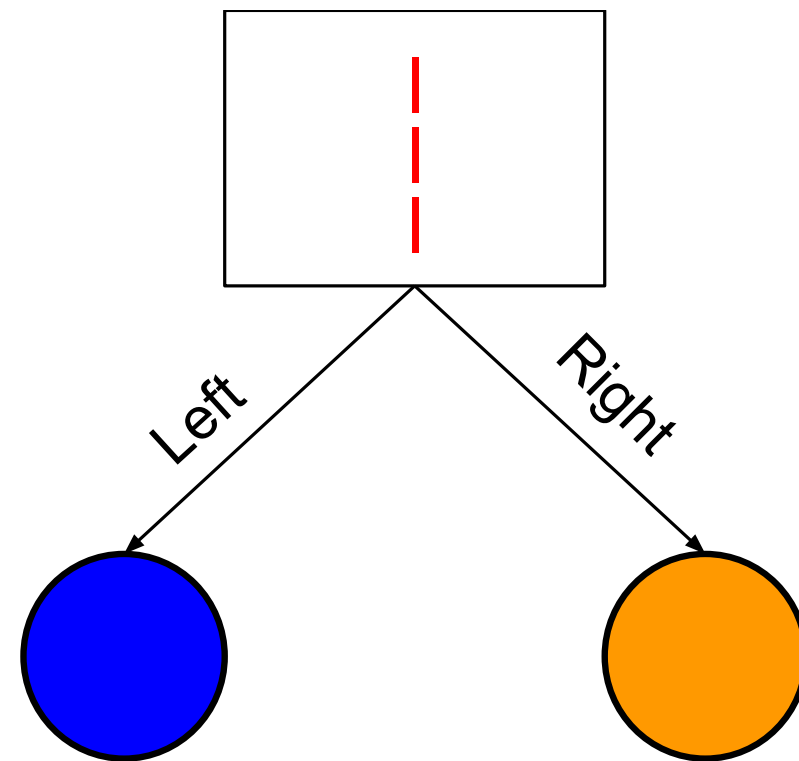


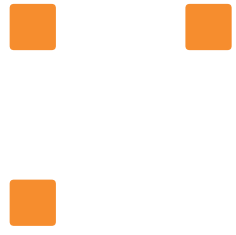
Growing a classification tree



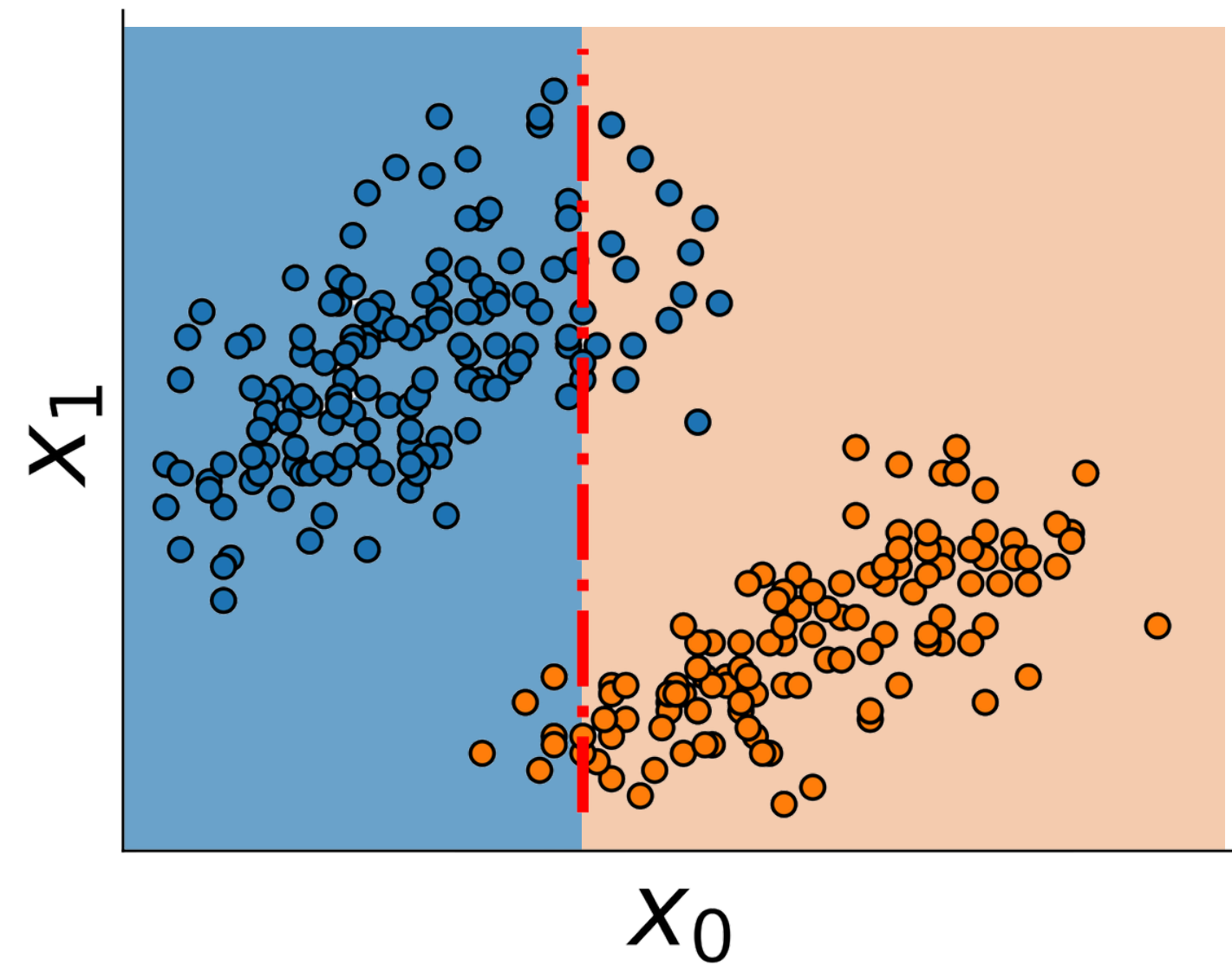
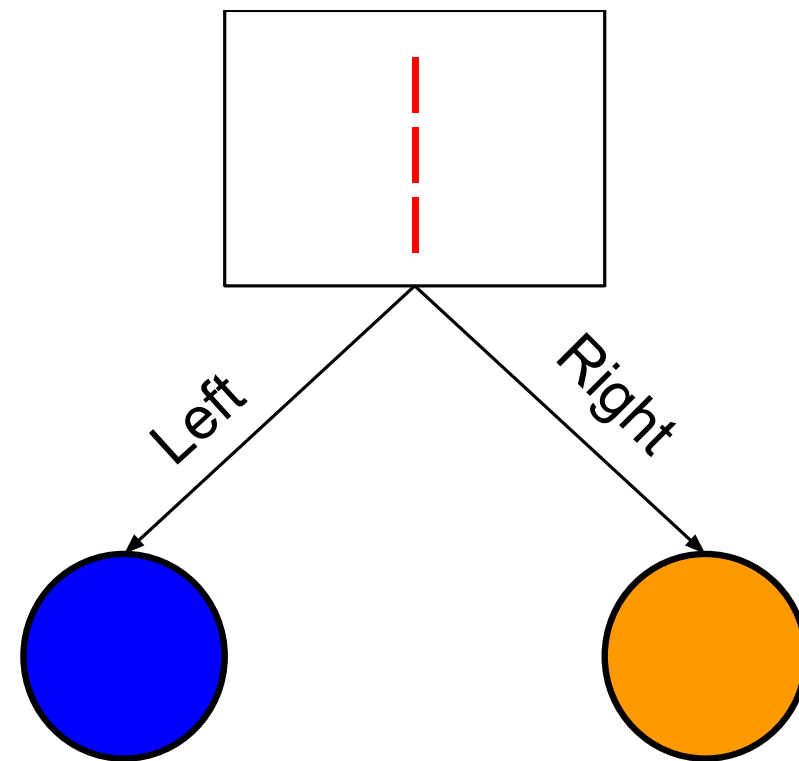


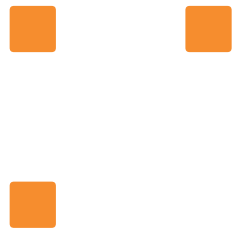
Growing a classification tree



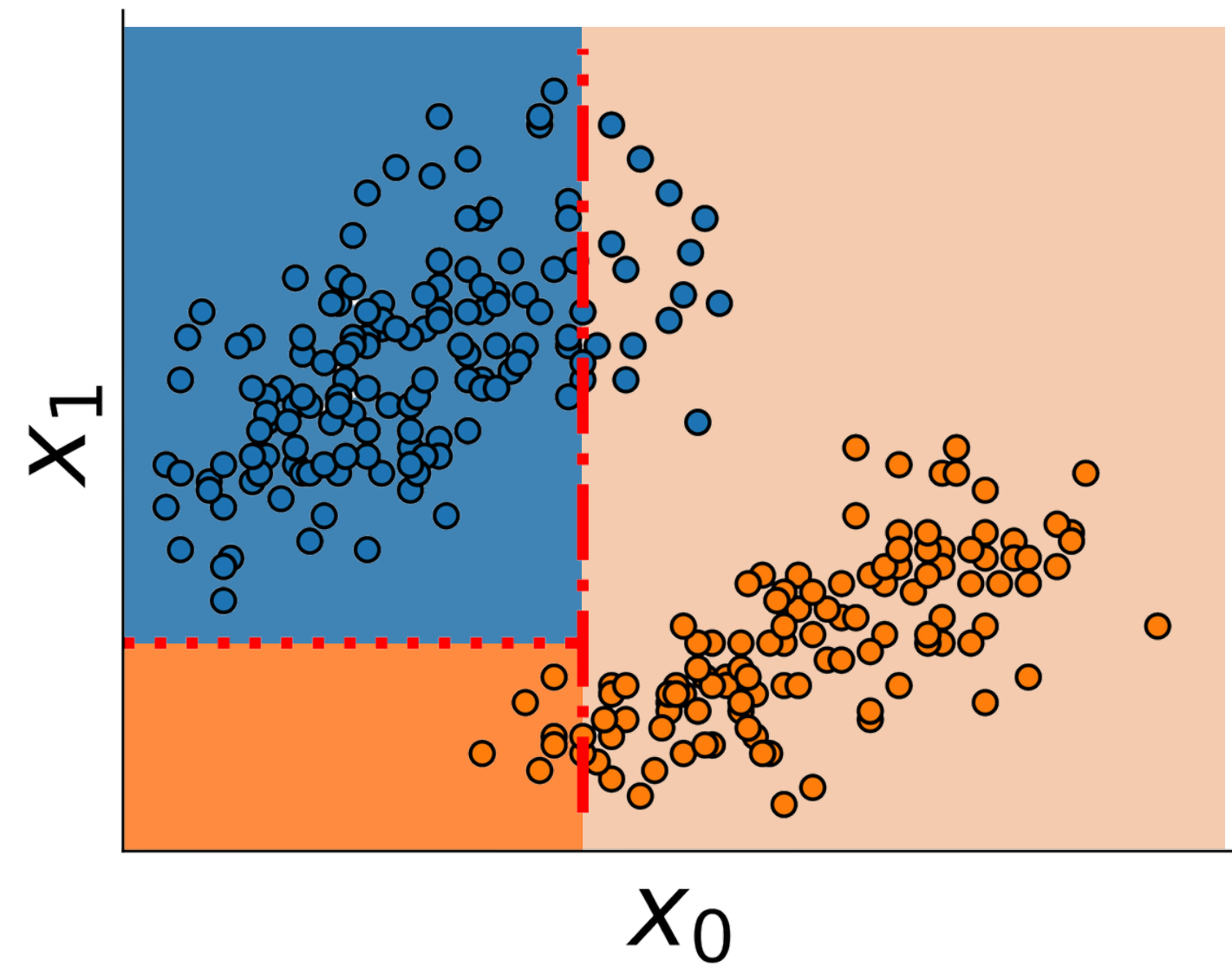
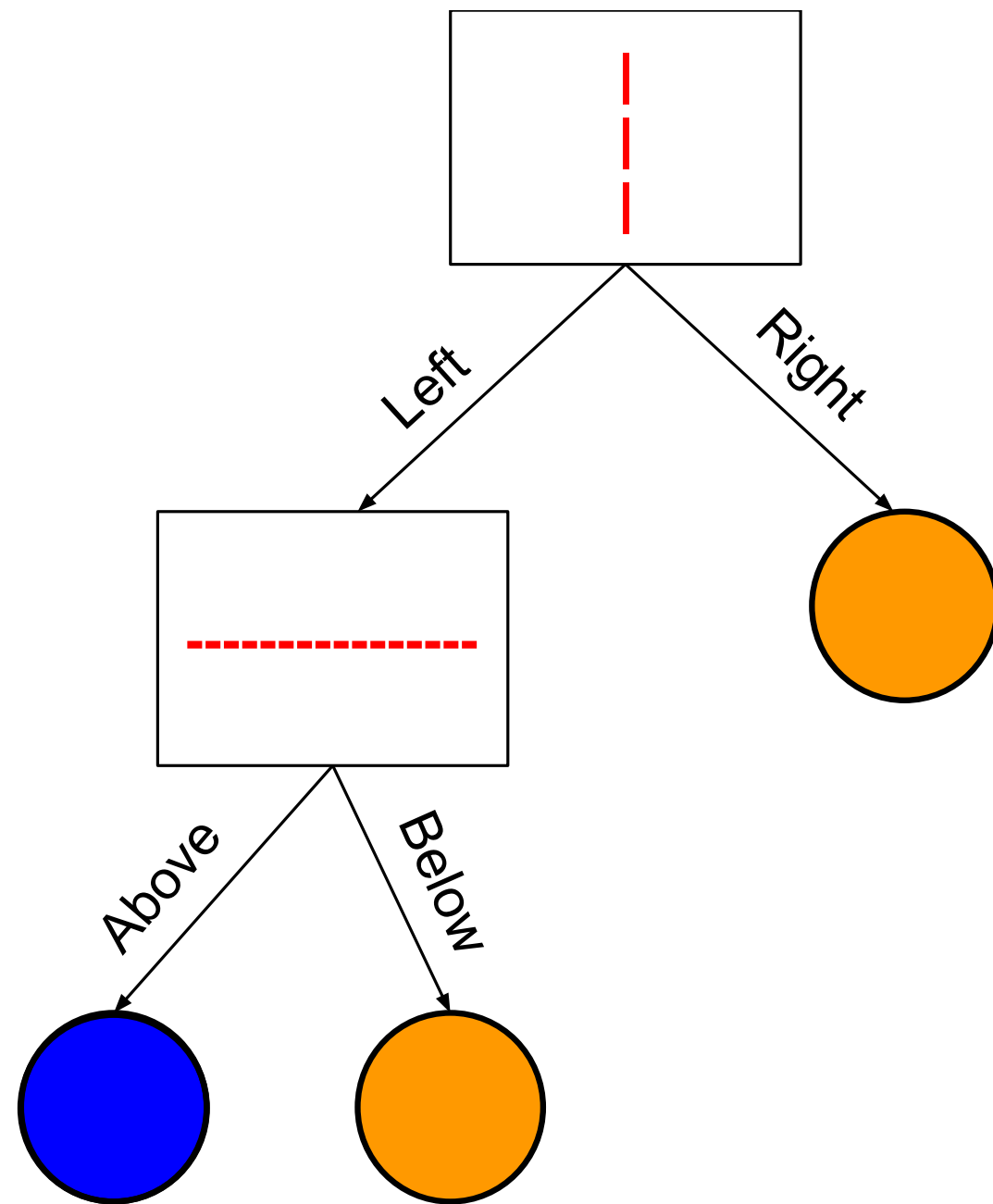


Growing a classification tree

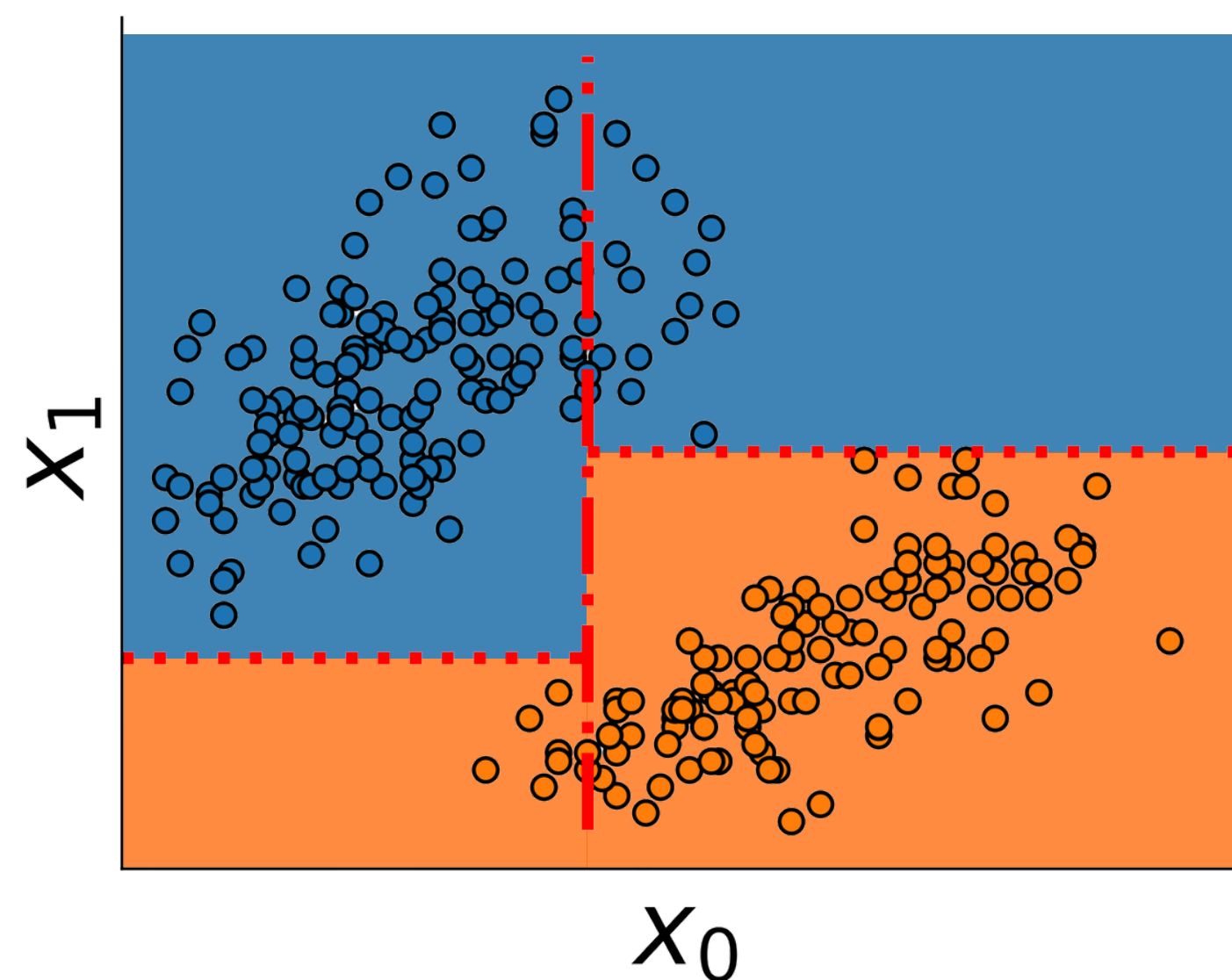
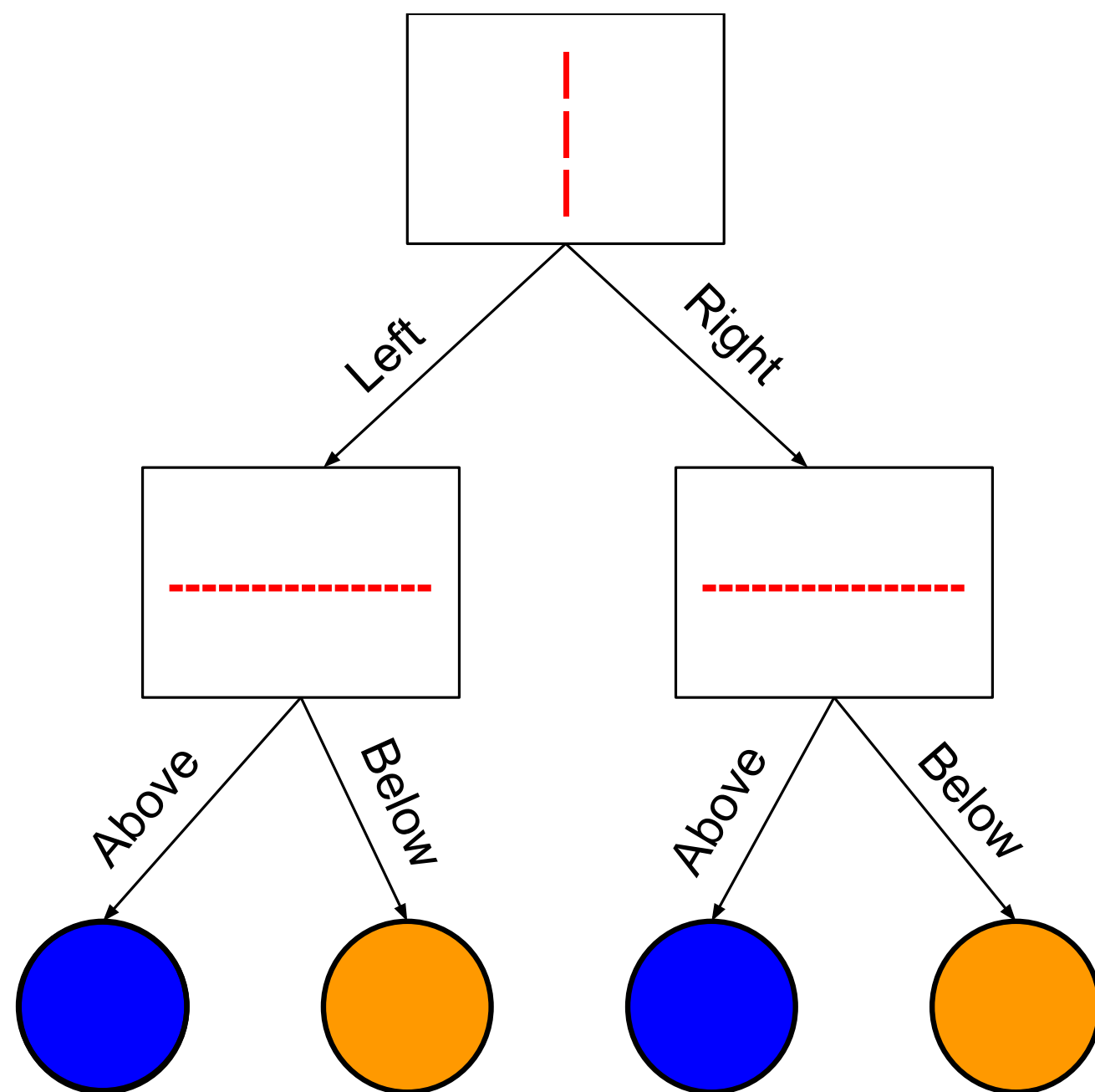


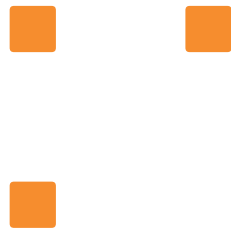


Growing a classification tree

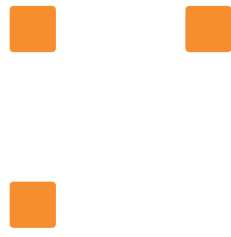


Growing a classification tree

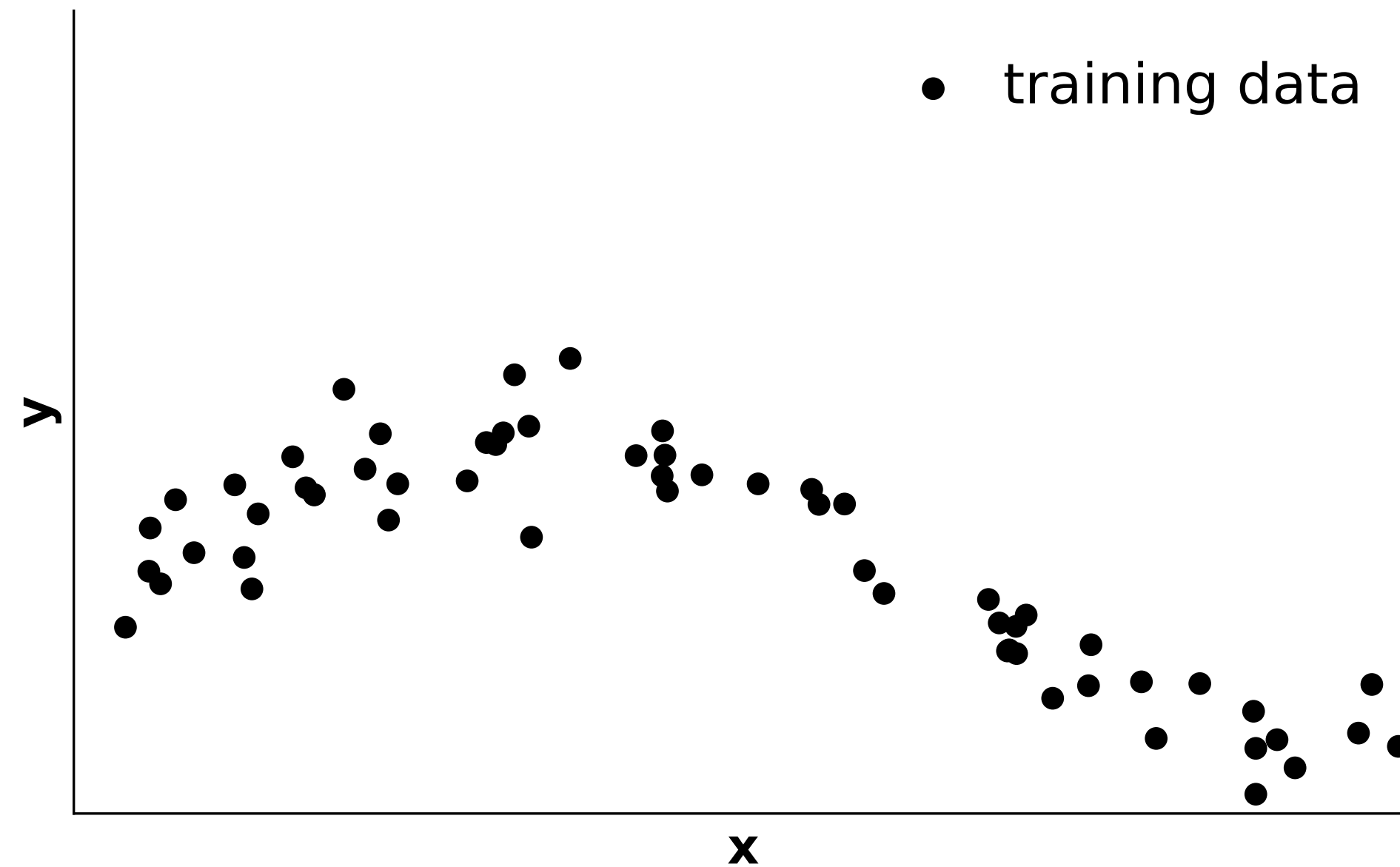


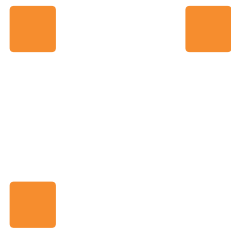


Decision trees in regression

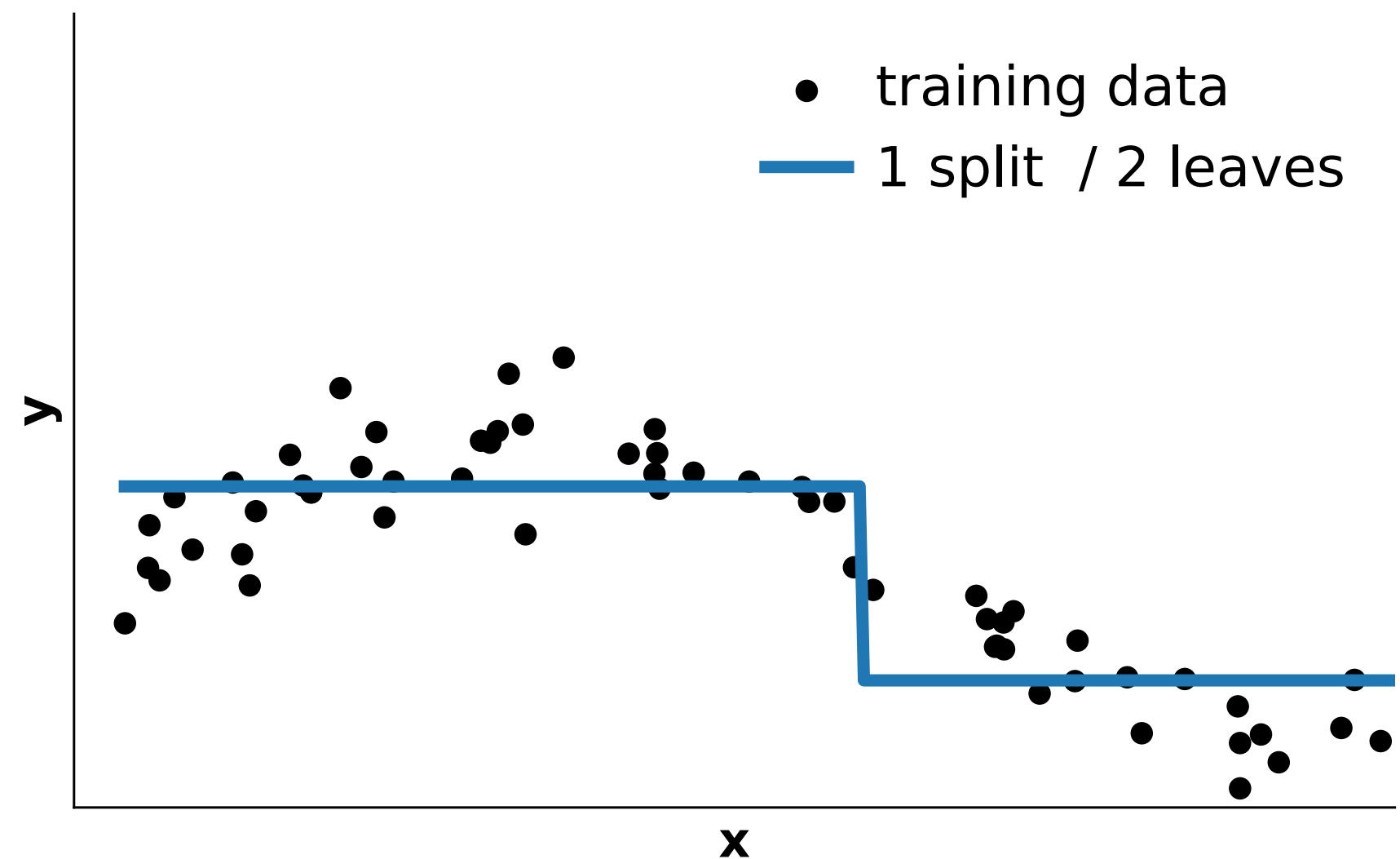
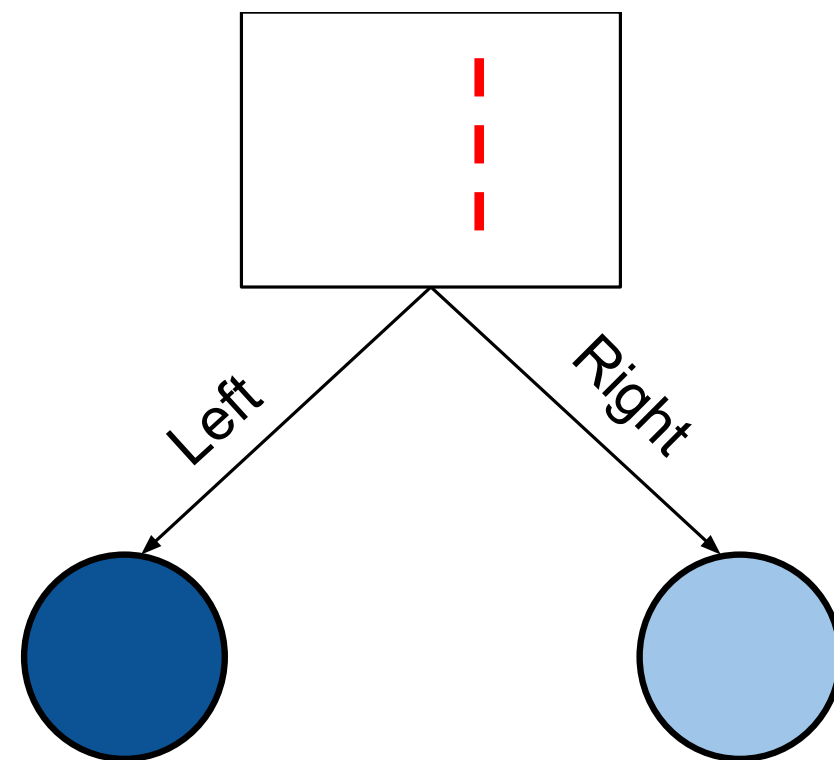


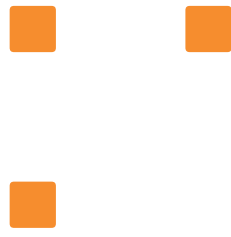
Growing a regression tree



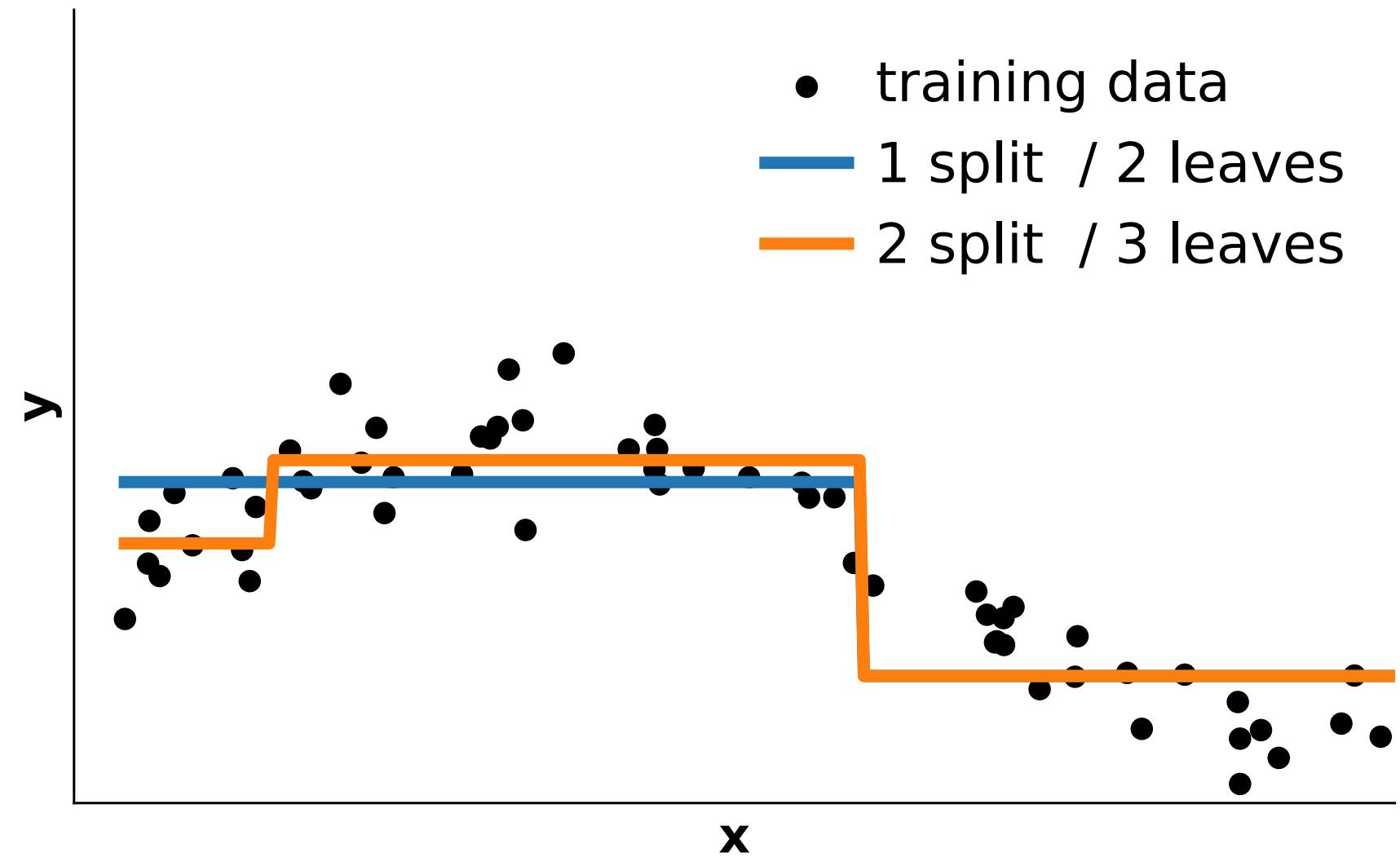
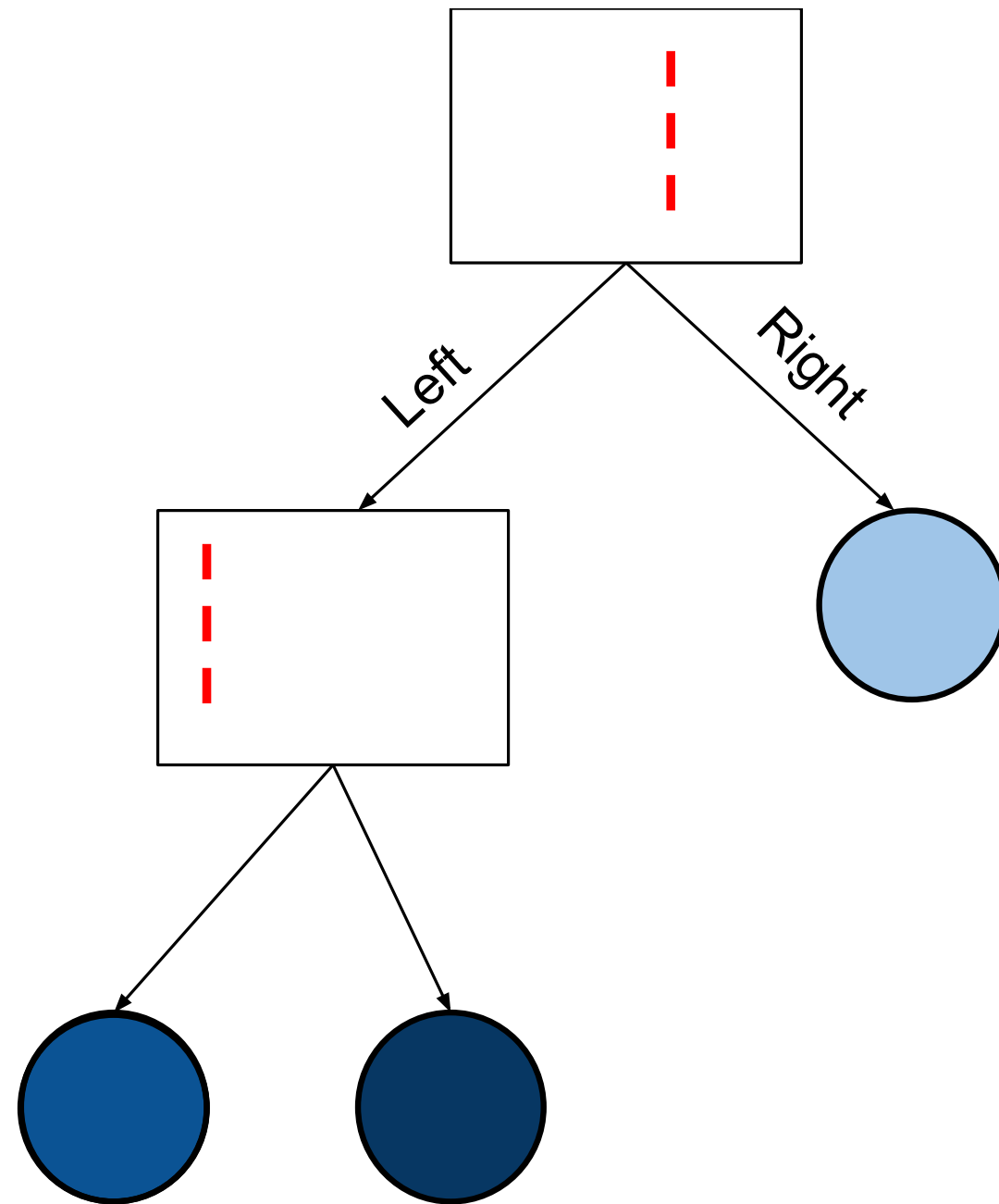


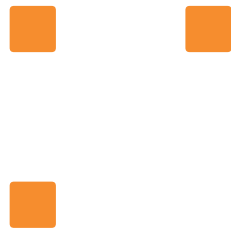
Growing a regression tree



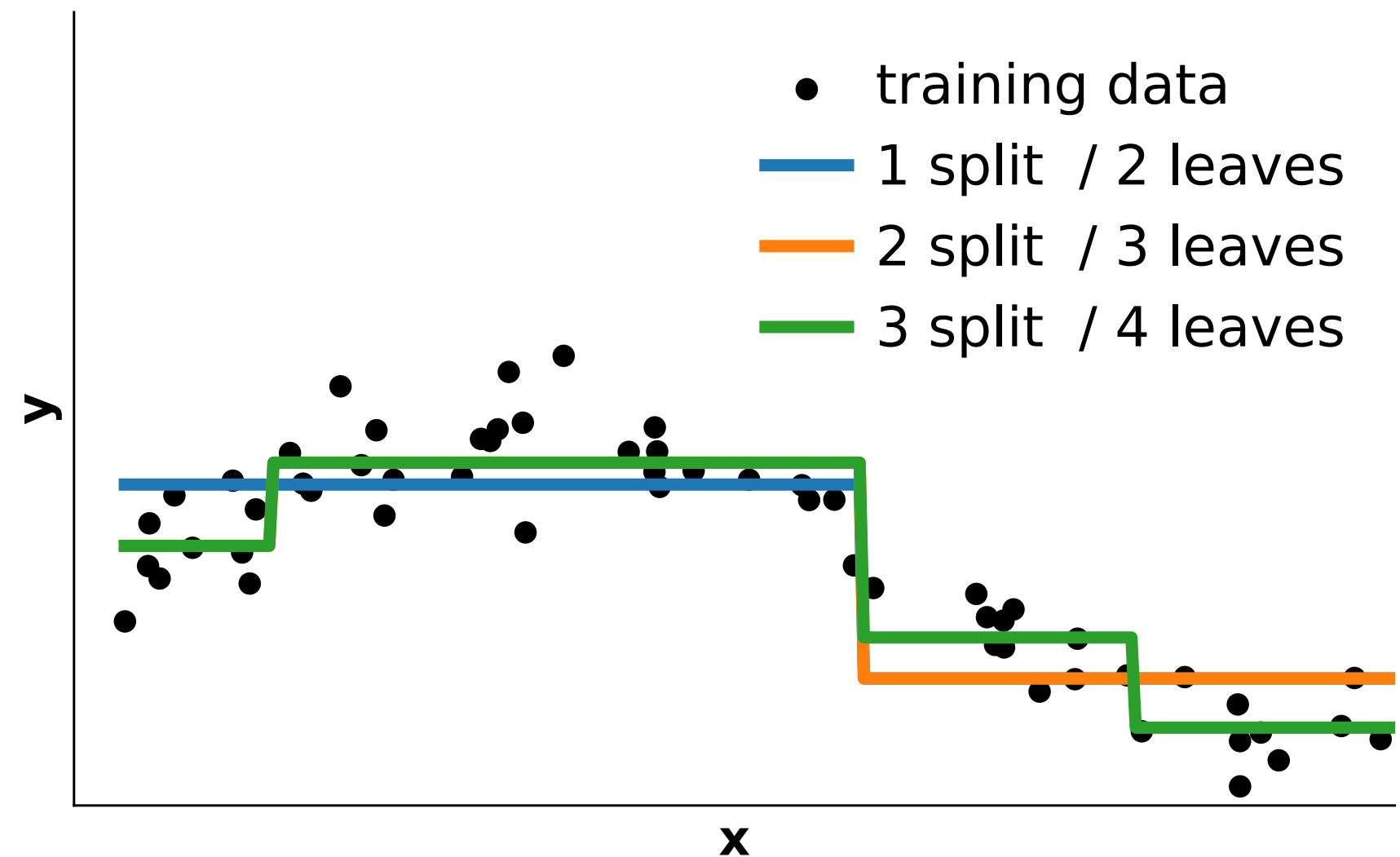
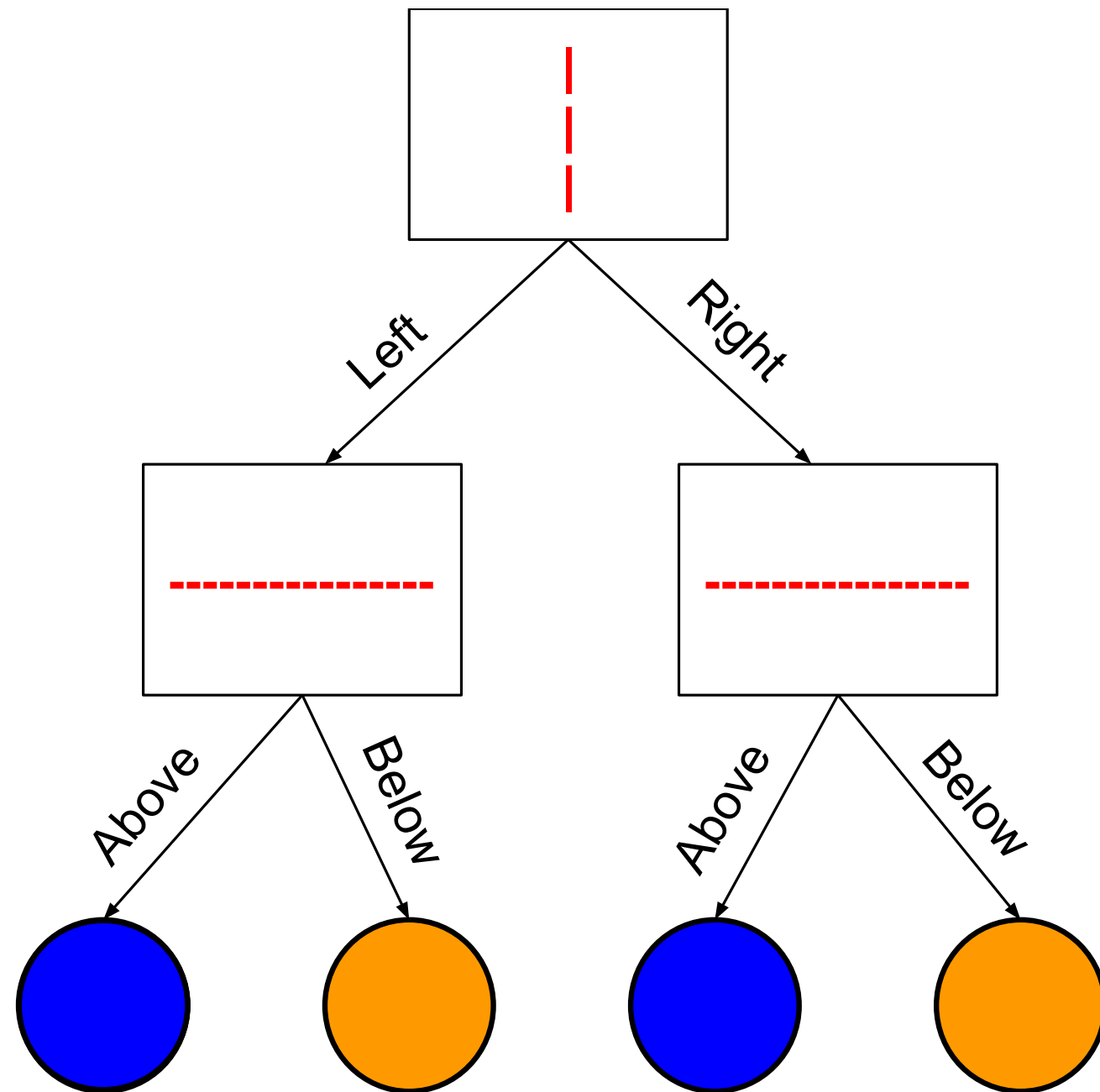


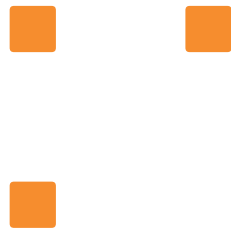
Growing a regression tree





Growing a regression tree

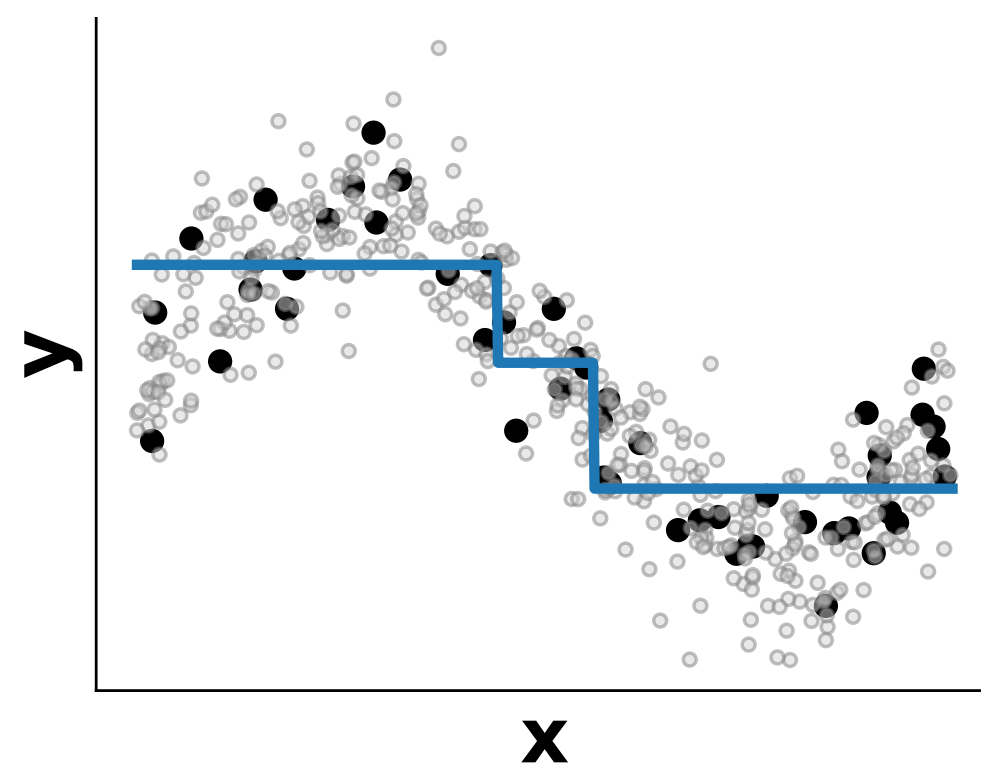




Tree depth and overfitting

How the maximum depth impacts generalization?

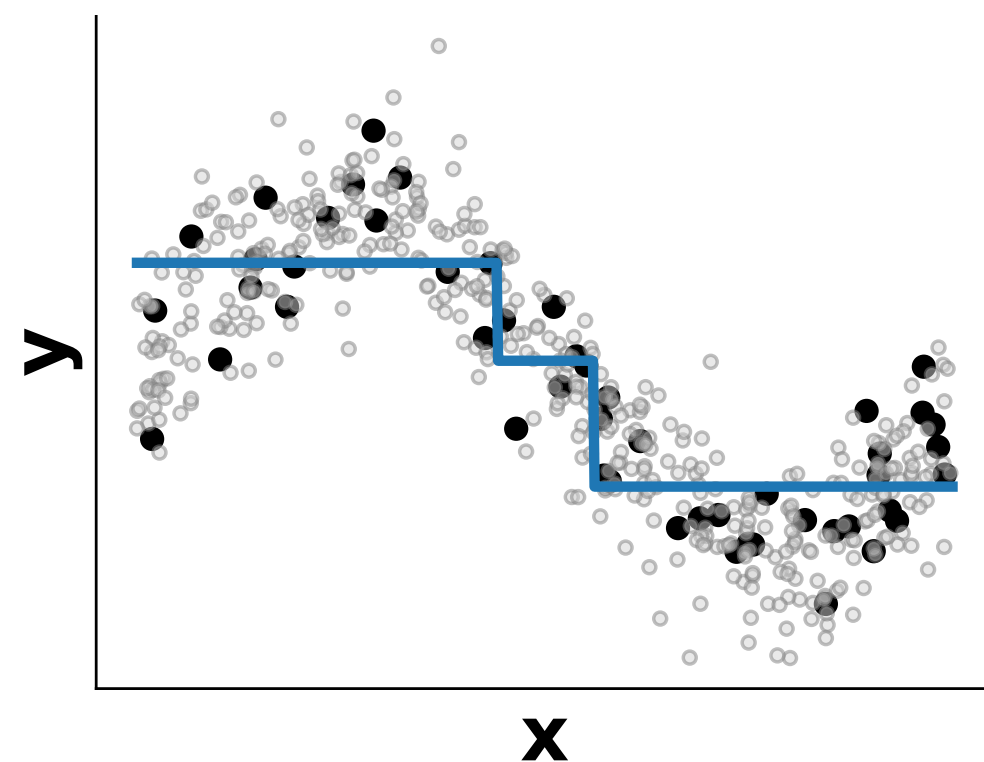
Tree: underfit / overfit tradeoff



Underfitting

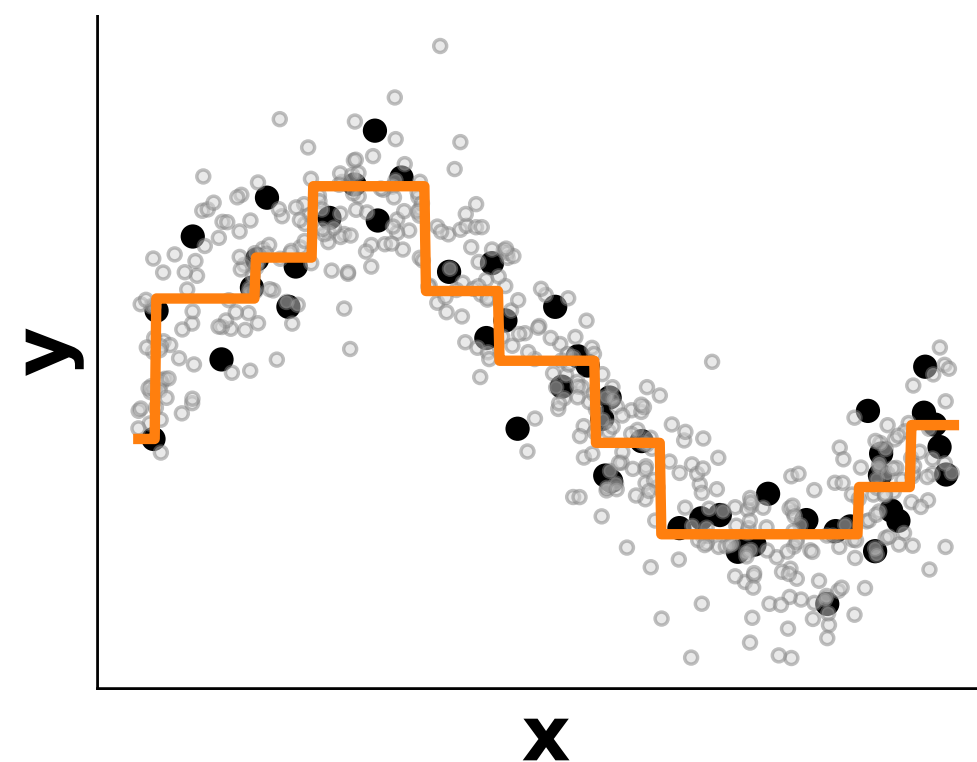
max_depth or
max_leaf_nodes
too small

Tree: underfit / overfit tradeoff



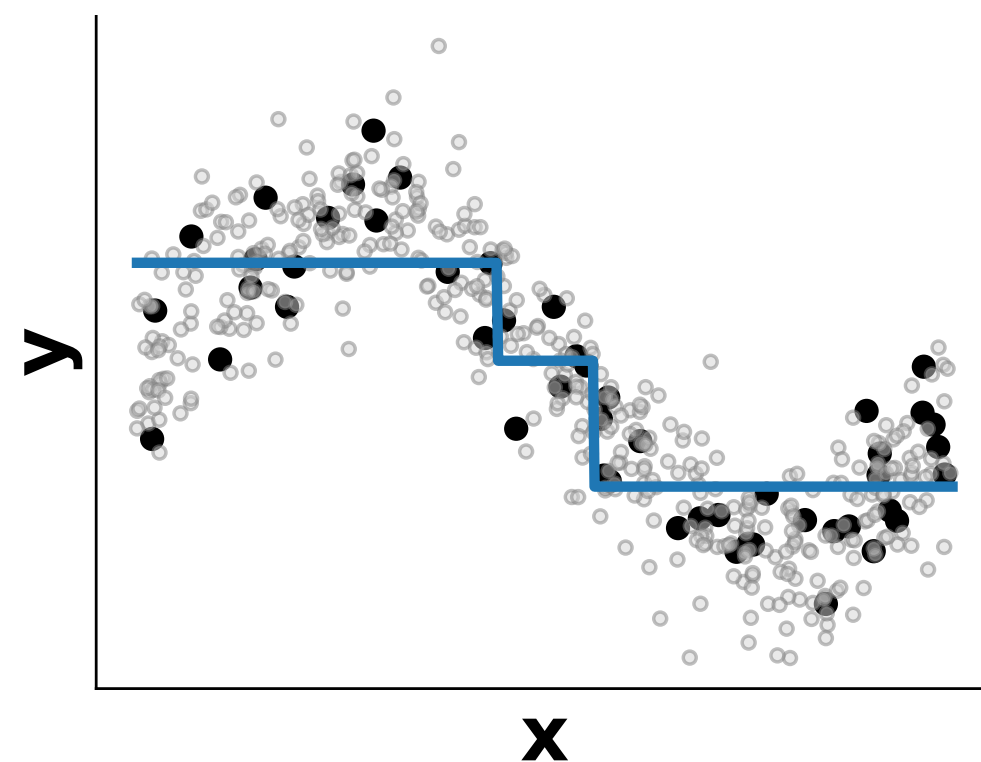
Underfitting

max_depth or
max_leaf_nodes
too small



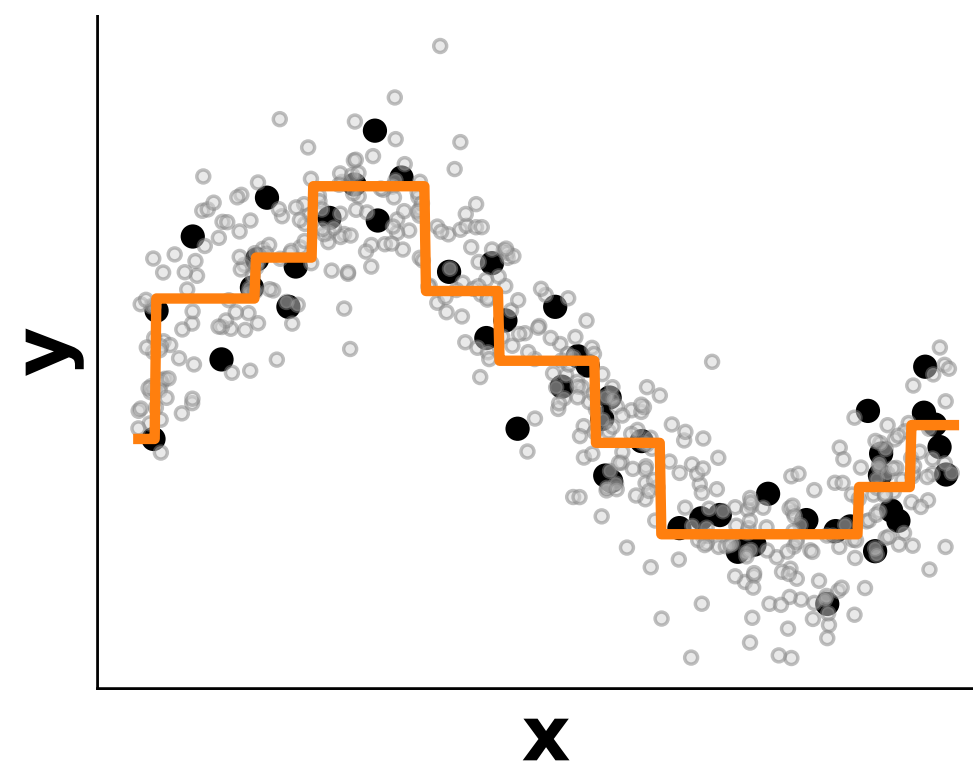
Best tradeoff

Tree: underfit / overfit tradeoff

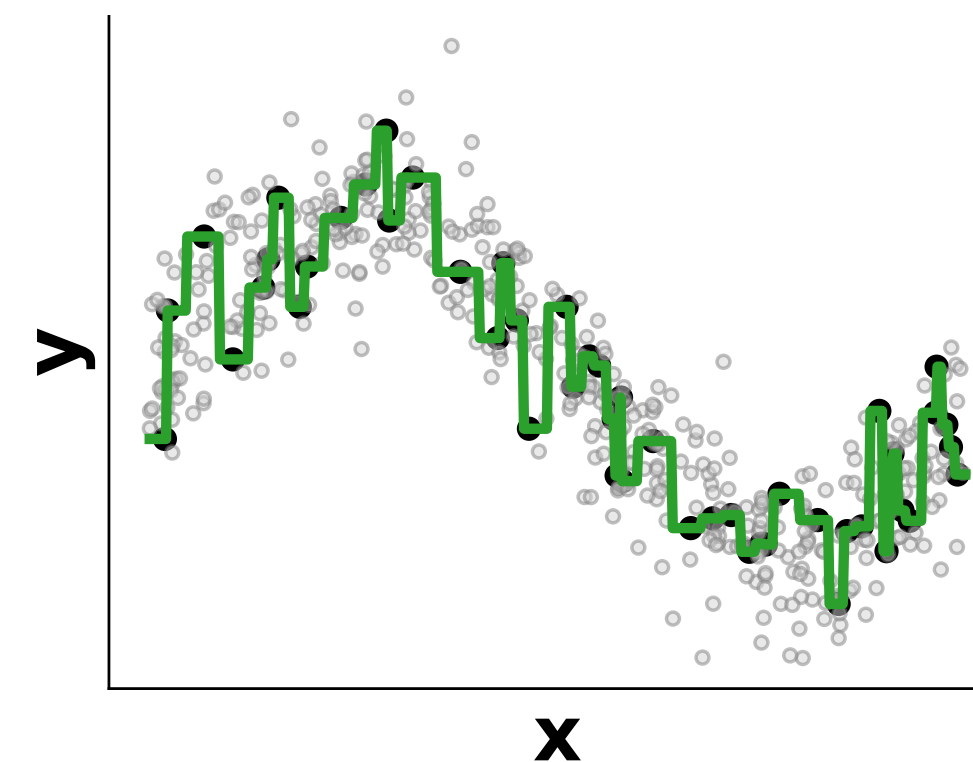


Underfitting

max_depth or
max_leaf_nodes
too small



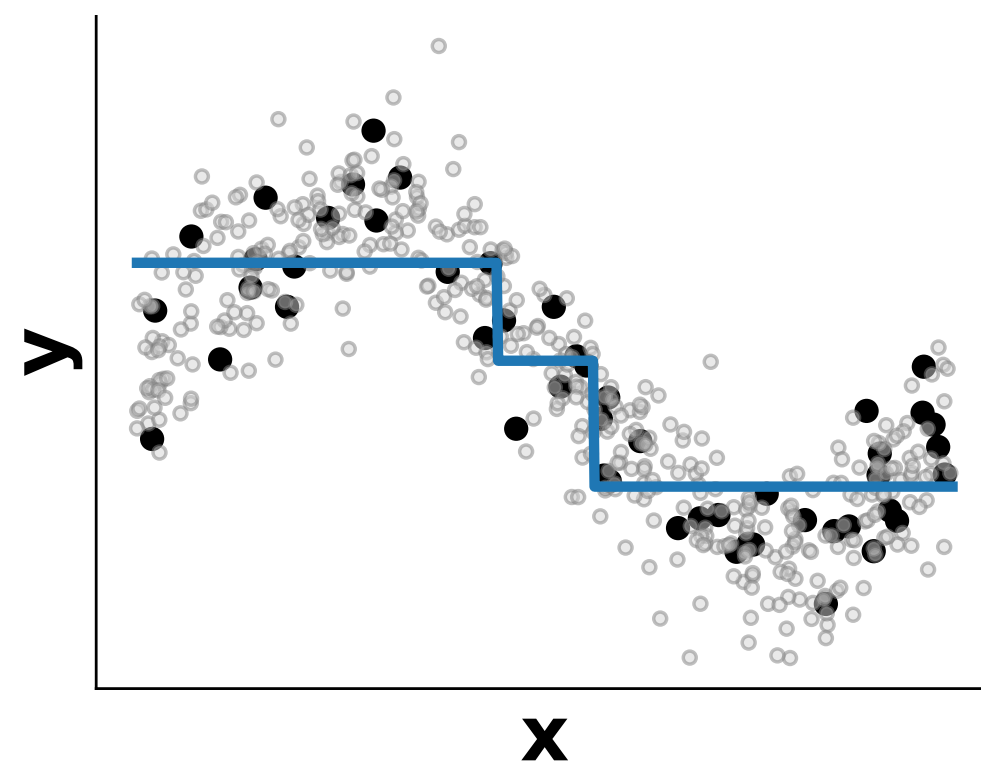
Best tradeoff



Overfitting

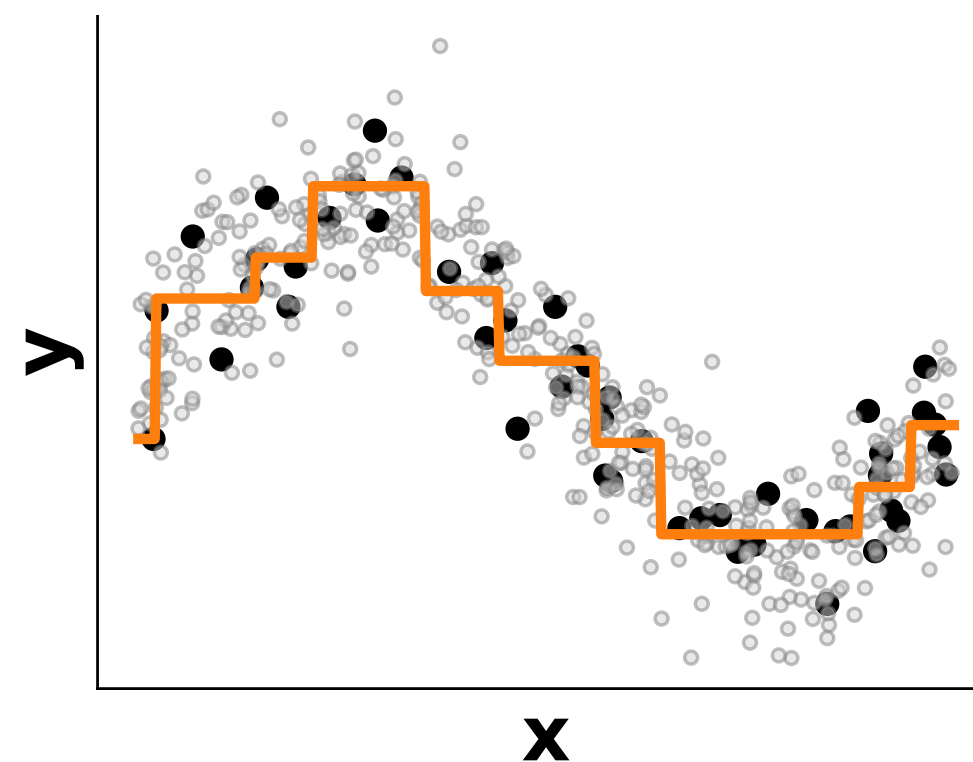
max_depth or
max_leaf_nodes
too large

Tree: underfit / overfit tradeoff

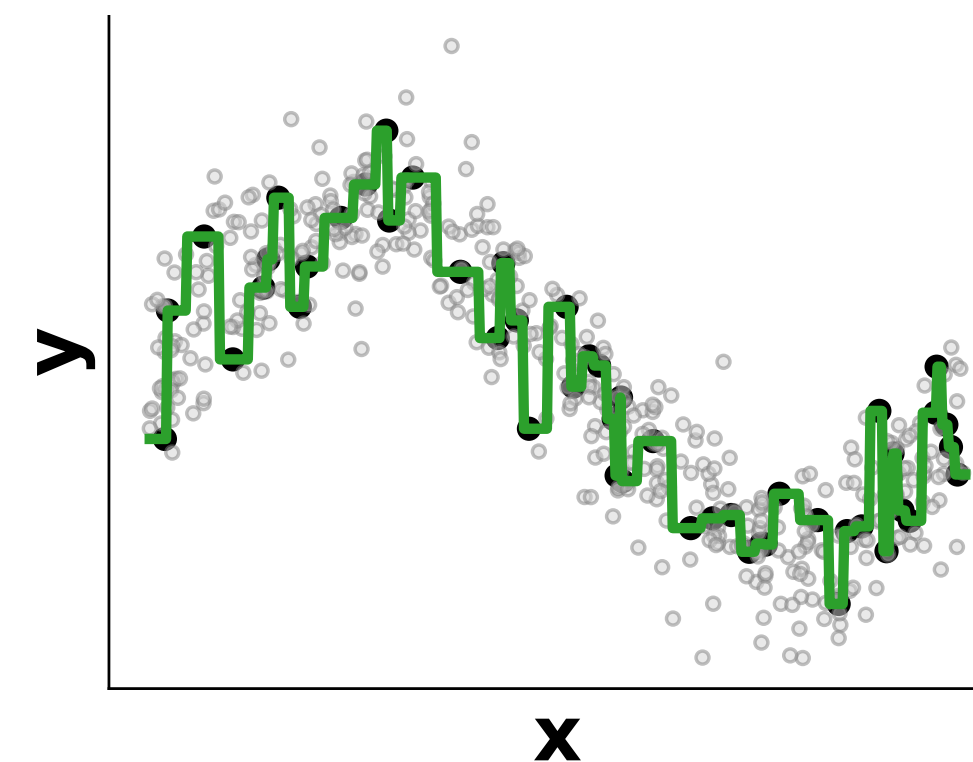


Underfitting

max_depth or
max_leaf_nodes
too small



Best tradeoff



Overfitting

max_depth or
max_leaf_nodes
too large

Take home messages

- Sequence of simple decision rules:
 - one feature and one threshold at a time
- `max_depth` and `max_leaf_nodes` control the trade-off between underfitting and overfitting
- No scaling required for numerical features
- Mostly useful as a building block for ensemble models
 - Random Forests
 - Gradient Boosting Decision Trees